

Challenge Overview

Event: PicoCTF 2019

Challenge Name: What Lies Within

Category: Forensics

Difficulty: Medium

Points: No points shown

Challenge Description

There's something in the building. Can you retrieve the flag?

Hint: There is data encoded somewhere... there might be an online decoder.

Step 1: Viewing the File

Download the file buildings.png to see what you can discover.



This is the buildings.png. As you can see nothing seems suspicious about this photo. It seems like an ordinary building.

Step 2: Digging Deeper into the Image

After viewing the image I remembered this part of the hint "There is data encoded somewhere." Exiftool is a phenomenal resource to discover encoded data. In a terminal use the command `exiftool buildings.png` to obtain additional information about the png.

```
L$ exiftool buildings.png

ExifTool Version Number      : 13.55
File Name                    : buildings.png
Directory                   : .
File Size                   : 625 kB
File Modification Date/Time  : 2026:08:13 18:56:31-07:00
File Access Date/Time       : 2026:08:13 18:56:34-07:00
File Inode Change Date/Time  : 2026:08:13 18:56:33-07:00
File Permissions             : -rw-rw-r--
File Type                   : PNG
File Type Extension         : png
MIME Type                   : image/png
Image Width                 : 657
Image Height                : 438
Bit Depth                  : 8
Color Type                  : RGB with Alpha
Compression                 : Deflate/Inflate
Filter                     : Adaptive
Interlace                   : Noninterlaced
Image Size                 : 657x438
Megapixels                  : 0.288
```

fgdf

This is what will appear when you use exiftool. As you can see nothing appears to be suspicious. time to try something else.

Step 3: Confirm its a PNG

Since we found nothing suspicious using exiftool let's use another tool to confirm that this is indeed a png and not something else or if something else is lurking within this png. We will use the file command to see if this truly is a png or not. The file command is part of the Linux utility that is utilized to identify the actual type and format of a file by examining its contents. The syntax for this command is *file buildings.png*.

```
L$ file buildings.png
buildings.png: PNG image data, 657 x 438, 8-bit/color RGBA, non-interlaced
```

As you can see the results from the file command confirms we are dealing with a png and nothing else. Time to try something else.

Step 4: Another Method

After using exiftool and confirming my findings with the file command I know for certain I'm dealing with a png and not any other type of extension. Looking back at the hint it states there's data encoded somewhere. With this knowledge it's time I thought of another tool that specifically works with extracting hidden data within a png and that tool is zsteg. I discovered zsteg a month prior to this challenge during another CTF challenge. Zsteg is a command-line steganography detection tool designed specifically for identifying and extracting hidden data embedded within png and bmp files. If you would like to know more about zsteg please visit the following site:

<https://linuxcommandlibrary.com/man/zsteg>. The syntax using zsteg for this challenge is *zsteg -a buildings.png*. The -a (or --all) flag for zsteg performs a comprehensive check across all known

steganography methods and channel combinations for any given png or bmp images. Let's see what happens when we use the command `zsteg -a buildings.png`.

[illegible]

Amazing! As soon as we use this command the flag is displayed instantaneously. Flag has been redacted in the screenshot above.

Closing Thoughts

In conclusion, the What Lies Within challenge was rewarding and fun. This challenge involved using different forensic techniques to solve this challenge which was phenomenal. As it builds the skills you will need in the cybersecurity environment. Even though I solved this challenge quickly I tried this challenge again and used other tools such as, binwalk, binwalk -e, identify -verbose, strings, etc. to see what else would happen. These other tools will definitely provide guidance on what your next steps should be to solve this challenge. What did you think of this challenge? What are other techniques you could use solve this challenge?

Securing the digital realm one computer at a time.

Best Regards,

Dylan